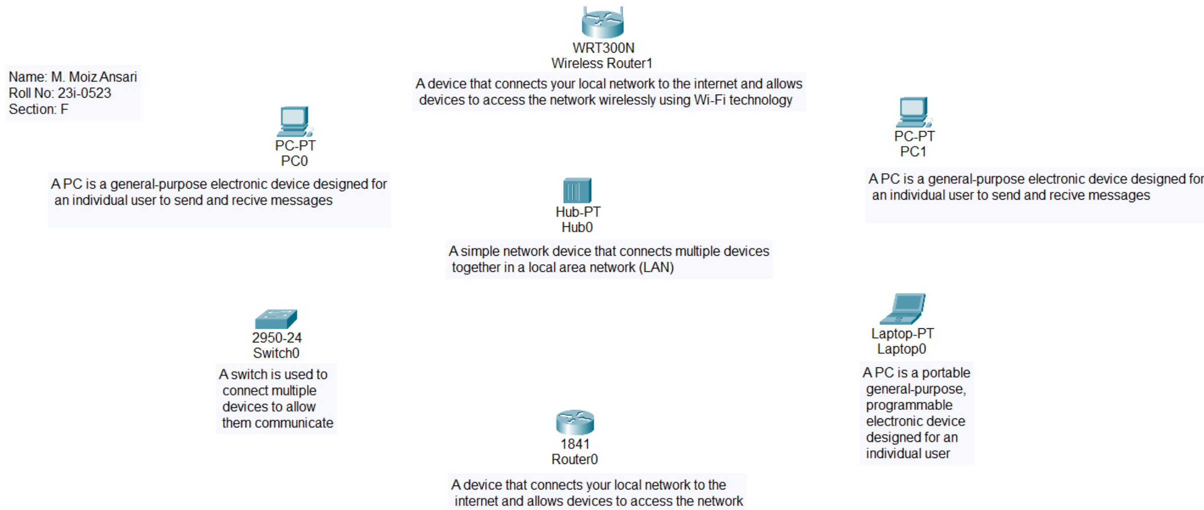
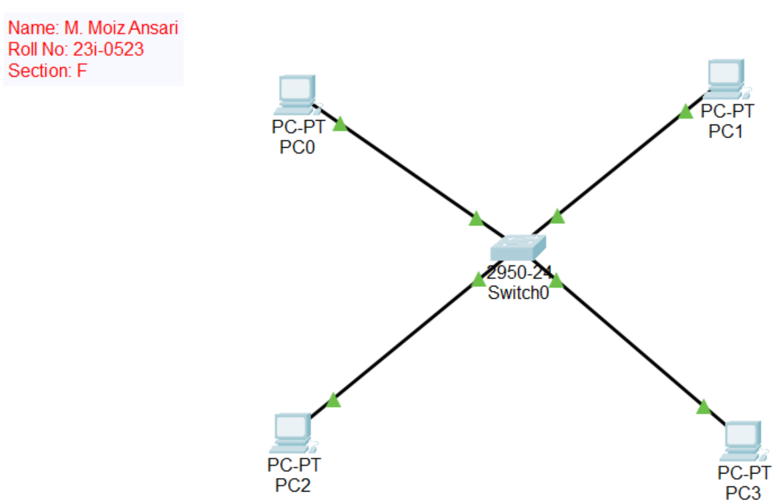
CNET Lab Task 2

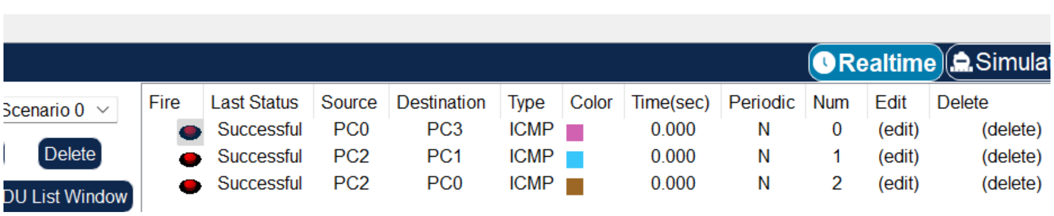
Task 1:



Task 2:

Star Topology:





Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC0	ICMP		0.000	N	2	(edit)	(delete)

In Unicast, the sender sends the message to the specific receiver, as it has Ips of all the tended receivers in a table. Whereas, in broadcast, sender sends message to all the tended receivers and only the receiver receives the message.

Advantage: Low latency, low cost

Disadvantage: If switch stops working, no alternate route to receiver

Ring Topology:

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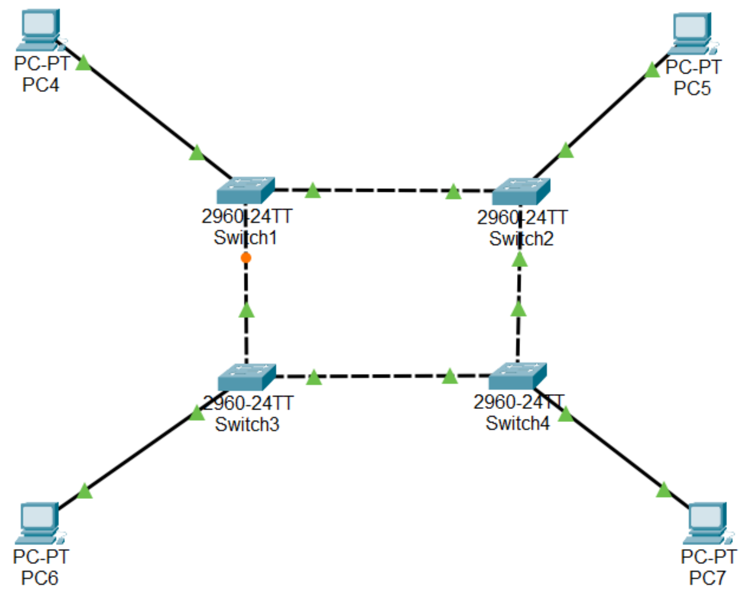


PC4

Physical Config Desktop Programming Attributes

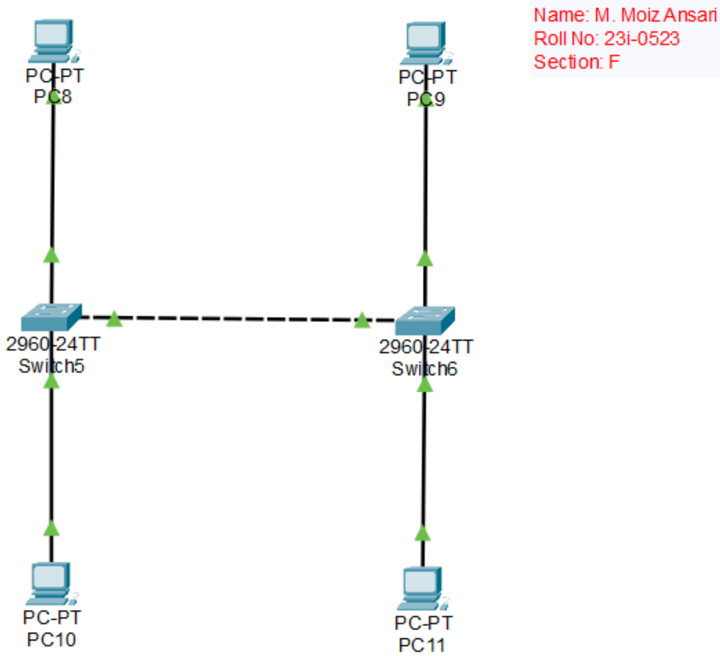
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time=29ms TTL=128

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 29ms, Average = 7ms
```

Advantage: More alternate route to receiver

Disadvantage: Higher latency than star topology

Bus Topology:



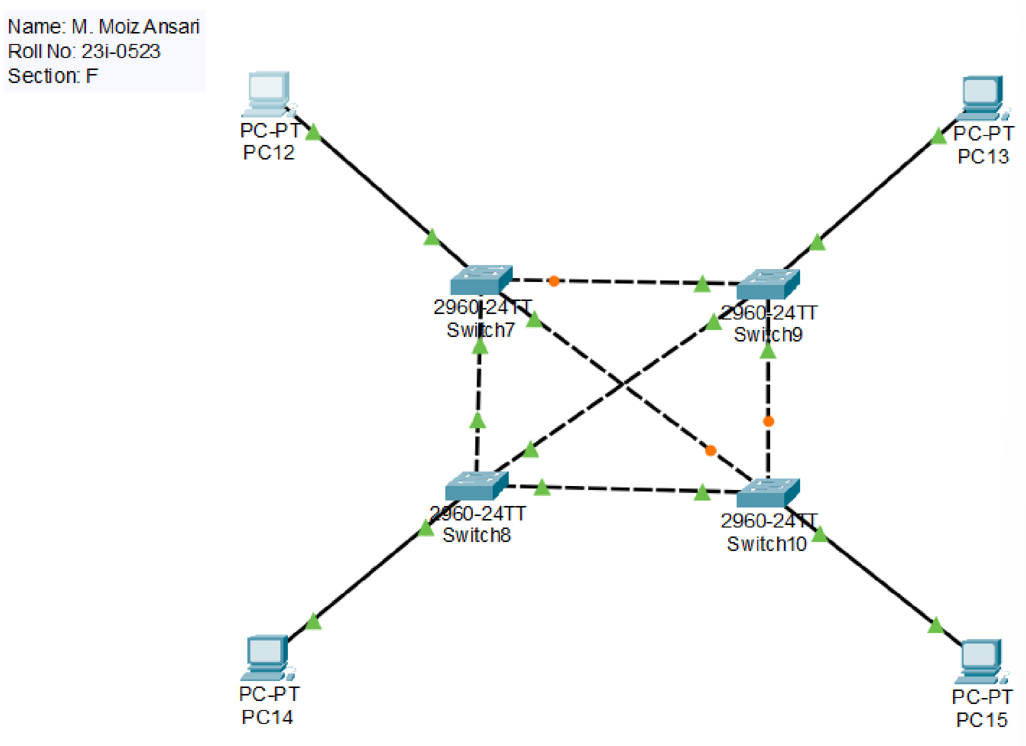
Command Prompt

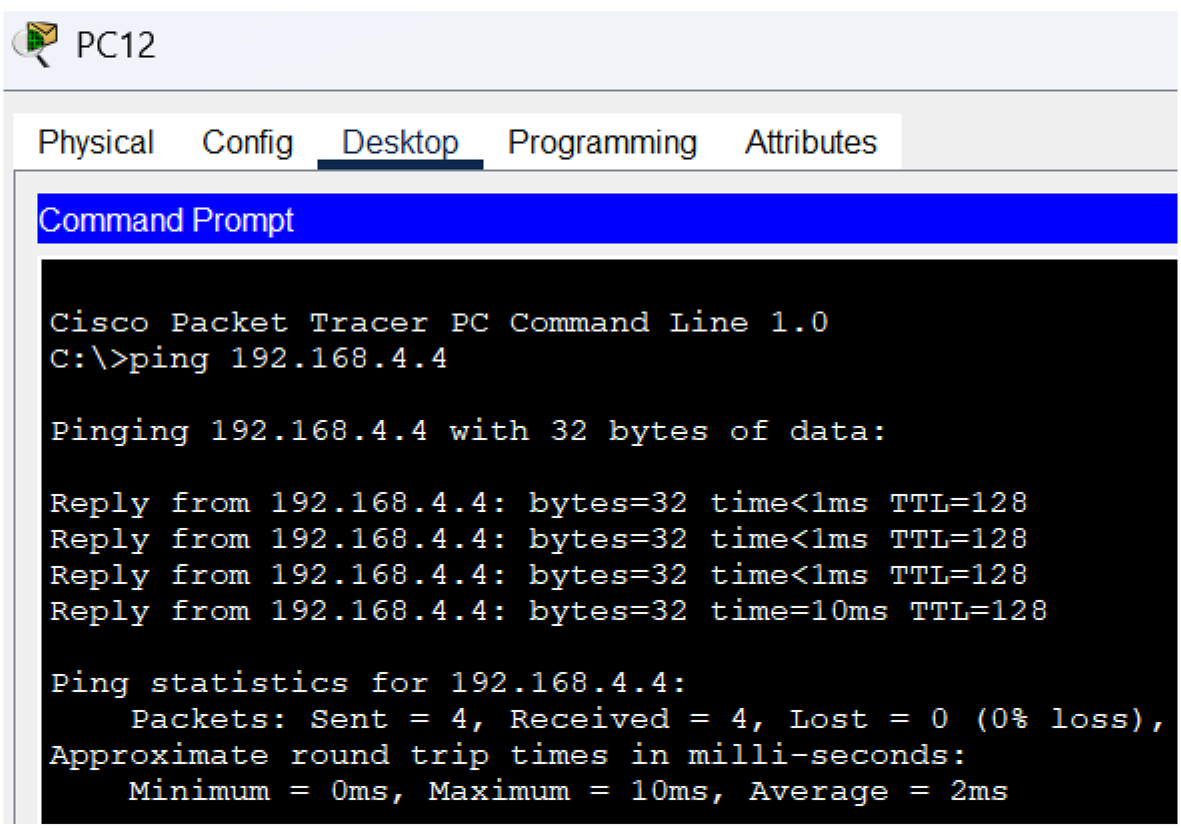
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.4

Pinging 192.168.3.4 with 32 bytes of data:

Reply from 192.168.3.4: bytes=32 time<1ms TTL=128
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128
Reply from 192.168.3.4: bytes=32 time=1ms TTL=128
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Advantage: Low cost, easy installation

Disadvantage: Reliance on a single cable for entire network

Mesh Topology:





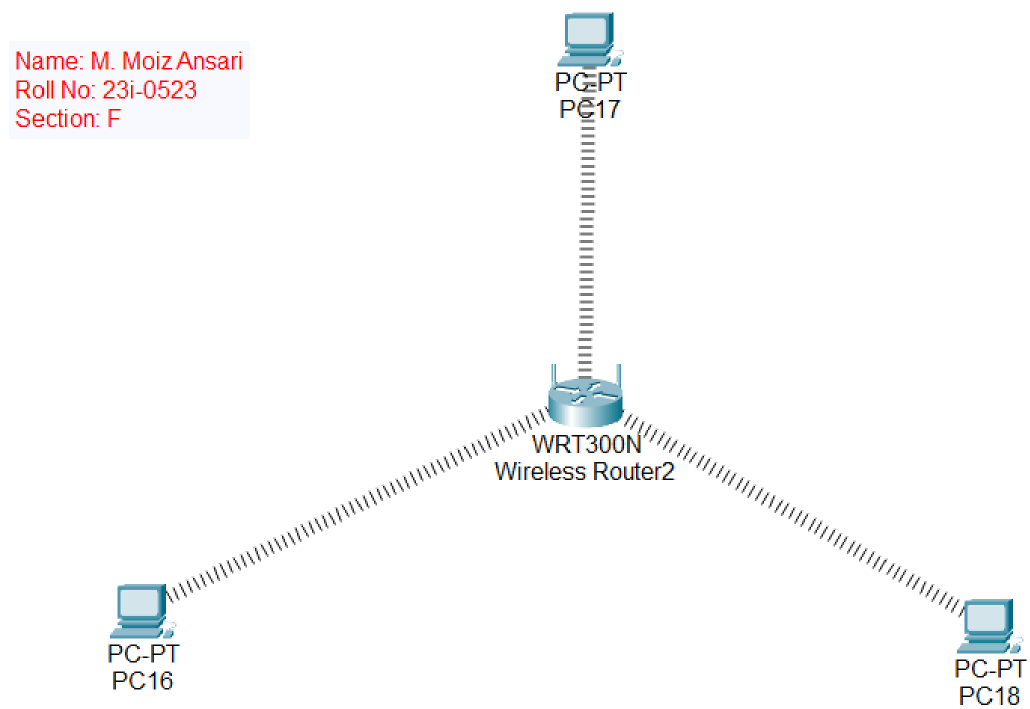

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.4.4

Pinging 192.168.4.4 with 32 bytes of data:

Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.4.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms
```

Mesh topology is redundant bcz it has multiple path from sender to receiver but its advantage is that if one path is broken, it can use the other path.

Advantage: Multiple paths to receiver

Disadvantage: High cost

Task 3:



PC1:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.0.11

Pinging 192.168.0.11 with 32 bytes of data:

Reply from 192.168.0.11: bytes=32 time=49ms TTL=128
Reply from 192.168.0.11: bytes=32 time=17ms TTL=128
Reply from 192.168.0.11: bytes=32 time=18ms TTL=128
Reply from 192.168.0.11: bytes=32 time=33ms TTL=128

Ping statistics for 192.168.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 49ms, Average = 29ms

C:\>ping 192.168.0.12

Pinging 192.168.0.12 with 32 bytes of data:

Reply from 192.168.0.12: bytes=32 time=53ms TTL=128
Reply from 192.168.0.12: bytes=32 time=36ms TTL=128
Reply from 192.168.0.12: bytes=32 time=25ms TTL=128
Reply from 192.168.0.12: bytes=32 time=28ms TTL=128

Ping statistics for 192.168.0.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 53ms, Average = 35ms

C:\>tracert 192.168.0.11

Tracing route to 192.168.0.11 over a maximum of 30 hops:

  1   23 ms     32 ms     20 ms     192.168.0.11

Trace complete.

C:\>tracert 192.168.0.12

Tracing route to 192.168.0.12 over a maximum of 30 hops:

  1   19 ms     26 ms     31 ms     192.168.0.12
```

```
Trace complete.

C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=48ms TTL=255
Reply from 192.168.0.1: bytes=32 time=24ms TTL=255
Reply from 192.168.0.1: bytes=32 time=19ms TTL=255
Reply from 192.168.0.1: bytes=32 time=27ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 48ms, Average = 29ms
```

PC2:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:

Reply from 192.168.0.10: bytes=32 time=35ms TTL=128
Reply from 192.168.0.10: bytes=32 time=23ms TTL=128
Reply from 192.168.0.10: bytes=32 time=23ms TTL=128
Reply from 192.168.0.10: bytes=32 time=23ms TTL=128

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 35ms, Average = 26ms

C:\>ping 192.168.0.12

Pinging 192.168.0.12 with 32 bytes of data:

Reply from 192.168.0.12: bytes=32 time=49ms TTL=128
Reply from 192.168.0.12: bytes=32 time=29ms TTL=128
Reply from 192.168.0.12: bytes=32 time=18ms TTL=128
Reply from 192.168.0.12: bytes=32 time=30ms TTL=128

Ping statistics for 192.168.0.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 49ms, Average = 31ms

C:\>tracert 192.168.0.10

Tracing route to 192.168.0.10 over a maximum of 30 hops:

  1   21 ms     20 ms     29 ms     192.168.0.10

Trace complete.

C:\>tracert 192.168.0.12

Tracing route to 192.168.0.12 over a maximum of 30 hops:

  1   22 ms     20 ms     14 ms     192.168.0.12
```

```
Trace complete.

C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=37ms TTL=255
Reply from 192.168.0.1: bytes=32 time=18ms TTL=255
Reply from 192.168.0.1: bytes=32 time=16ms TTL=255
Reply from 192.168.0.1: bytes=32 time=8ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 37ms, Average = 19ms
```

PC3:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
ping 192.268.0.10
Ping request could not find host 192.268.0.10. Please check the name and try again.
C:\>ping 192.168.0.10

Pinging 192.168.0.10 with 32 bytes of data:

Reply from 192.168.0.10: bytes=32 time=25ms TTL=128
Reply from 192.168.0.10: bytes=32 time=23ms TTL=128
Reply from 192.168.0.10: bytes=32 time=18ms TTL=128
Reply from 192.168.0.10: bytes=32 time=17ms TTL=128

Ping statistics for 192.168.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 17ms, Maximum = 25ms, Average = 20ms

C:\>ping 192.168.0.11

Pinging 192.168.0.11 with 32 bytes of data:

Reply from 192.168.0.11: bytes=32 time=32ms TTL=128
Reply from 192.168.0.11: bytes=32 time=34ms TTL=128
Reply from 192.168.0.11: bytes=32 time=27ms TTL=128
Reply from 192.168.0.11: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 34ms, Average = 26ms

C:\>tracert 192.168.0.10

Tracing route to 192.168.0.10 over a maximum of 30 hops:

  1   33 ms     9 ms      25 ms     192.168.0.10

Trace complete.

C:\>tracert 192.168.0.11

Tracing route to 192.168.0.11 over a maximum of 30 hops:
```

```
C:\>tracert 192.168.0.11

Tracing route to 192.168.0.11 over a maximum of 30 hops:

  1   19 ms     28 ms     22 ms     192.168.0.11

Trace complete.

C:\>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=53ms TTL=255
Reply from 192.168.0.1: bytes=32 time=20ms TTL=255
Reply from 192.168.0.1: bytes=32 time=11ms TTL=255
Reply from 192.168.0.1: bytes=32 time=21ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 53ms, Average = 26ms
```